Scaling Laws for a Genotype-to-Phenotype Model

The two published forms, what counts as N, D and C when the examples are strains rather than tokens, and the smallest ablation that gives a curve worth extrapolating

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Abstract

A scaling law is a fitted statement that held-out loss falls as a power of model size N, training data D or training compute C, and its value is that a curve fit on runs one can afford predicts runs one cannot. Two forms carry the literature. Kaplan et al. fit each axis alone, $L = (N_c/N)^{\alpha_N}$ and $L = (D_c/D)^{\alpha_D}$, straight lines on log-log axes. Hoffmann et al. fit the joint form $L(N, D) = E + AN^{-\alpha} + BD^{-\beta}$ with an irreducible floor E, and minimizing it under the transformer budget $C \approx 6ND$ gives the compute-optimal allocation $N^* \propto C^a$, $D^* \propto C^b$ with $a + b = 1$. One figure draws both forms at their published constants, the plateaus each axis runs into when the other is fixed, the IsoFLOP curves whose minima trace the frontier, and a synthetic fitting exercise that holds out the largest runs. For genotype-to-phenotype data the questions that decide whether the curve means anything are what to count: D is the number of training examples the optimizer consumes in the unit the loss averages over, not the dimension of an example; repeated epochs count for less than fresh data; the evaluation set is fixed and split by the unit that leaks, a gene or a perturbation, not by record; and E has a physical reading, the replicate-noise floor, which the released replicate standard deviations estimate independently of any fit. No torchcell run has been fit to either form; the last section is the plan for doing so on the trigenic build, where data, not parameters, is the axis that binds.

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1 The two forms ✗

A **scaling law** is a fitted relation between a model's held-out loss and one or more of the three quantities that a training run spends: the number of trainable parameters N, the amount of training data D, and the training compute C. The claim behind the name is that, over the range a family of models was trained on, the loss falls as a power of each quantity, so that a line fit on log-log axes to the runs one can afford predicts the loss of a run one cannot. The literature carries two functional forms, and the difference between them is a floor.

Kaplan: one axis at a time. Kaplan et al. (arXiv:2001.08361, in the mirror) fit each axis separately, with the other axes made large enough not to limit:

$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha_N}, \qquad L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D}. \tag{1}$$

Here L is the **test loss**, the average loss on a fixed held-out evaluation set, cross-entropy per token for a language model. N_c and D_c are fitted constants in units of parameters and data; they set where each curve sits and have no meaning of their own. α_N and α_D are the **scaling exponents**, the slopes on log-log axes, and the larger one names the axis along which a decade of spending buys more. For language models Kaplan reports $\alpha_N \approx 0.076$ and $\alpha_D \approx 0.095$. Compute enters through the transformer rule

$$C \approx 6ND \text{ FLOPs}, \tag{2}$$

about two floating-point operations per parameter per training token for the forward pass and four for the backward pass. N in that rule and in Equation 1 counts *non-embedding* parameters; including the embedding matrix bends the curve at small N, because the embedding scales with the vocabulary rather than with the capacity the loss responds to.

Chinchilla: a joint form with a floor. Hoffmann et al. (arXiv:2203.15556, in the mirror) fit both axes at once and let the loss bottom out:

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}. \tag{3}$$

E is the **irreducible loss**, the value approached with infinite parameters and infinite data; it comes from noise or inherent unpredictability in the data, the entropy of text for a language model. A and B are the coefficients of the model-size and data terms, and α, β are the exponents of the joint fit. Their Approach 3 fit gives $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha = 0.34$, $\beta = 0.28$ on their training runs. With a floor present, a pure power law is wrong at the large end: on log-log axes L bends toward E while $L - E$ stays straight, which is the diagnostic panel c of Figure 1 draws.

Compute-optimal allocation. Minimizing Equation 3 subject to the budget $C = 6ND$ gives the size and data a fixed budget should buy. Writing $D = C/(6N)$ and setting the derivative in N to zero,

$$N^*(C) = G\left(\frac{C}{6}\right)^{a}, \qquad D^*(C) = \frac{C}{6N^*(C)}, \qquad G = \left(\frac{\alpha A}{\beta B}\right)^{\frac{1}{\alpha+\beta}}, \qquad a = \frac{\beta}{\alpha+\beta}, \quad b = \frac{\alpha}{\alpha+\beta}, \tag{4}$$

so $N^* \propto C^a$ and $D^* \propto C^b$ with $a + b = 1$. The Hoffmann constants give $a = 0.452$ and $b = 0.548$, parameters and data growing at nearly the same rate; Kaplan's earlier conclusion that model size should grow much faster than data came from learning-rate schedules not matched to each run's length, which Hoffmann corrected. The **compute-optimal frontier** is the curve $(N^*(C), D^*(C))$ in the (N, D) plane, and an **IsoFLOP curve** is the loss along one budget line $6ND = C$, a U in N whose minimum sits on the frontier. SRC: `experiments/041-scaling-laws/scripts/scaling_law_forms.py`, `ChinchillaFit.a / .b / .G`

The same runs, a different fit. Besiroglu et al. (arXiv:2404.10102, in the mirror) refit the Approach 3 data and got $E = 1.82$, $A = 482$, $B = 2085$, $\alpha = 0.35$, $\beta = 0.37$, which moves the allocation to $a = 0.513$ and $b = 0.487$. The same runs support two frontiers that differ visibly over the budgets that matter (panel h), which is the standing argument for reporting an interval on every exponent rather than a point.

Symbols. Three names in the fitting code of Section 4 clash with the symbols above and are listed here so the clash is visible. Table 2 is hand-authored.

Table 1: Published constants of the two forms, and the allocation exponents they imply. Kaplan's N_c and D_c set the height of a pure power law and have no meaning on their own; Hoffmann's and Besiroglu's rows are two fits of the same Chinchilla training runs, and the difference between them is the point of panel h. Every value is from outside our holdings.

fit	symbol	value	units
Kaplan 2020	N_c	8.8×10^{13}	parameters
	α_N	0.076	–
	D_c	5.4×10^{13}	tokens
	α_D	0.095	–
Hoffmann 2022	E	1.69	loss
	A	406.4	loss · parameters$^{\alpha}$
	B	410.7	loss · tokens$^{\beta}$
	α	0.34	–
	β	0.28	–
	$a = \beta/(\alpha+\beta)$	0.452	derived
	$b = \alpha/(\alpha+\beta)$	0.548	derived
Besiroglu 2024	E	1.8172	loss
	A	482.01	loss · parameters$^{\alpha}$
	B	2085.43	loss · tokens$^{\beta}$
	α	0.3478	–
	β	0.3658	–
	$a = \beta/(\alpha+\beta)$	0.513	derived
	$b = \alpha/(\alpha+\beta)$	0.487	derived

Table 2: Every symbol in Equations 1 to 4 and in the estimator, with the three names that collide.

symbol	meaning
L	test loss: average loss on a fixed, separate held-out set
N	trainable parameters; Kaplan counts non-embedding parameters only
D	training data the optimizer consumed, in the unit the loss averages over (tokens, or records here)
C	training compute in floating-point operations
6	FLOPs per parameter per training example for a transformer, forward plus backward
N_c, D_c	Kaplan's fitted scale constants, units of parameters and of data
α_N, α_D	Kaplan's single-axis exponents
E	irreducible loss, the floor
A, B	coefficients of the finite-size and finite-data terms
α, β	exponents of the joint fit
N^*, D^*	compute-optimal size and data for a budget C
a, b	allocation exponents, $N^* \propto C^a$, $D^* \propto C^b$, $a + b = 1$
G	prefactor of the optimum, Equation 4
`log_a`, `log_e`	in the estimator: $\log A$ and $\log E$, fit in log space so both stay positive; not the allocation exponents a, b
`alpha`	in the estimator: the joint-fit α
`r`	the residual, predicted log-loss minus observed log-loss
`starts`	the grid of initial guesses the optimizer is run from, because the floored fit has several local minima

2 Reading the figure ✗

Figure 1 draws both forms at the published constants of Table 1. Every panel is the function, not a measurement; the one panel with points, panel i, draws runs from a known curve to show the fitting recipe on data whose answer is known. The rows go from one axis at a time, to the joint surface, to what the surface implies for spending.

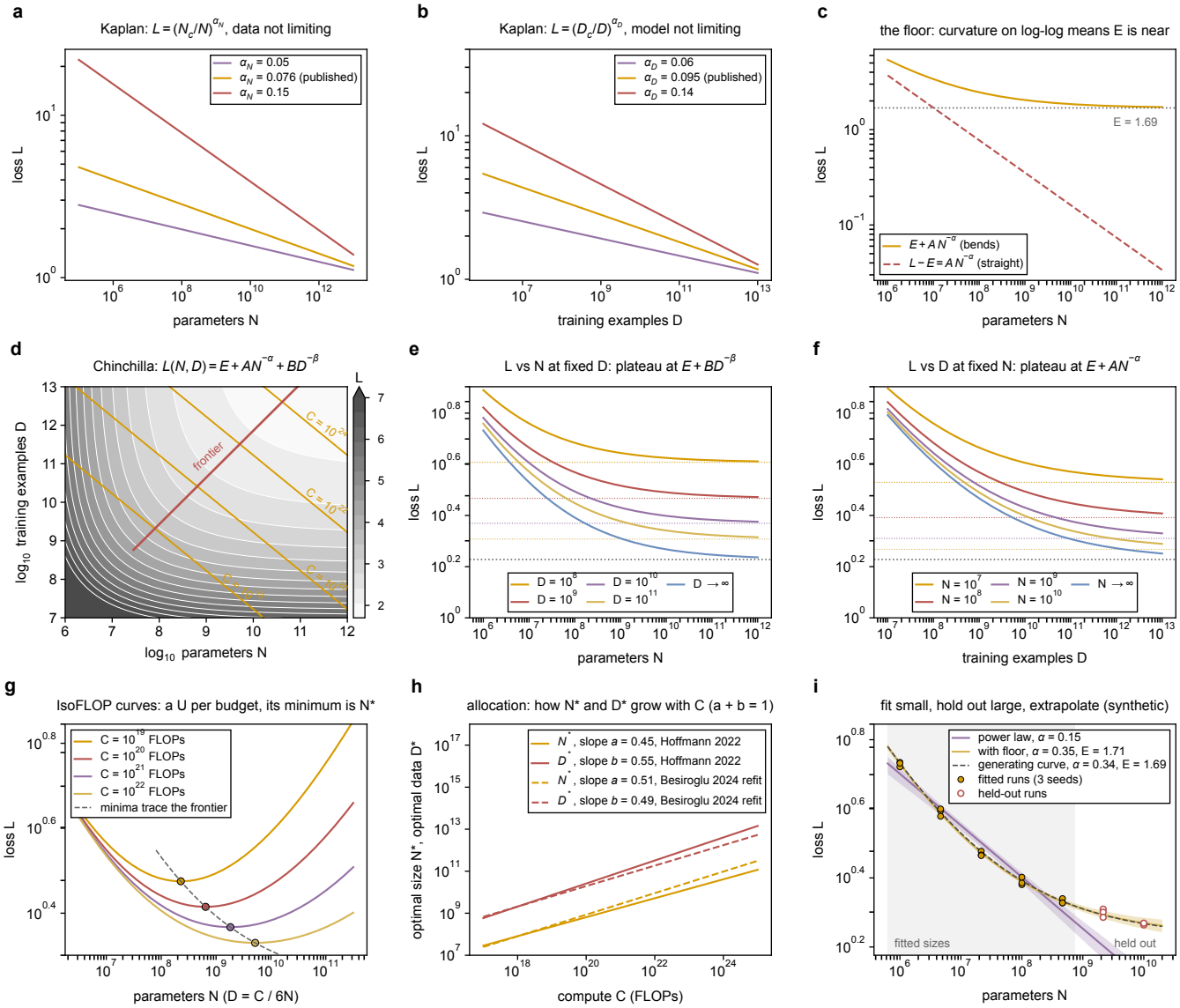


Figure 1: The two scaling-law forms and what follows from them. **a, b**, Kaplan's single-axis laws, Equation 1, with the published exponent and one smaller and one larger. On log-log axes the exponent is the slope, and every curve passes through $L = 1$ at $N = N_c$ or $D = D_c$. **c**, the irreducible floor: $E + AN^{-\alpha}$ at the Hoffmann constants bends toward E (dotted) while $L - E$ stays a straight line. **d**, the joint form, Equation 3, as a surface over $\log_{10} N$ and $\log_{10} D$; the amber diagonals are lines of constant compute $6ND = C$ and the brick line is the compute-optimal frontier, where each diagonal touches its lowest contour. **e**, cuts of the surface at fixed D: more parameters stop paying once the curve reaches the plateau $E + BD^{-\beta}$ (dotted, per curve). **f**, cuts at fixed N: more data stops paying at $E + AN^{-\alpha}$. **g**, IsoFLOP curves, the loss along one diagonal of d, drawn against N with $D = C/(6N)$; each U has a minimum (marked) at $N^*(C)$, and the minima lie on the frontier (dashed). **h**, the allocation, Equation 4: N^* and D^* against C on log-log axes, under the Hoffmann fit (solid) and the Besiroglu refit of the same runs (dashed); the slopes are a and b. **i**, a synthetic fitting exercise. Fifteen runs, three seeds at each of five sizes from 10^6 to $10^{8.67}$, are drawn from $E + AN^{-\alpha}$ at the Hoffmann constants with 2% multiplicative noise, and two sizes above are held out. A pure power law (lilac) and a floored power law (wheat) are fit on the fifteen and extrapolated, with 5 to 95% bootstrap bands over runs. The generating curve is dashed.

SRC: `experiments/041-scaling-laws/scripts/scaling_law_forms.py`

What to look for in each row. In the top row the exponent is everything: panel a shows that a change of α_N from 0.076 to 0.15 moves the loss predicted at $N = 10^{13}$ by less than it moves the loss at 10^5, since every curve is pinned at N_c, but an exponent fit at small N and used at large N moves the forecast by the full difference in slope times the decades extrapolated. Panel c is the diagnostic to run before trusting any straight-line fit: if the largest runs sit below the line, the floor has been reached or something else has become the bottleneck, and the curve to fit is $L - E$.

In the middle row, every cut plateaus, and the height of the plateau is set by the axis held fixed. Panel e is the picture of a data-limited project: at $D = 10^8$ examples the Hoffmann surface flattens at $L = 4.1$ no matter how many parameters are added, and reaching $L = 3$ needs ten times the data before it needs a larger model. Panel f is the mirror image. The dotted plateau lines are $E + BD^{-\beta}$ and $E + AN^{-\alpha}$, and the gap between a plateau and E is what the other axis can still buy.

In the bottom row the question is spending. The IsoFLOP minima in panel g sit at $N^* = 2.3 \times 10^8$, 6.4×10^8, 1.8×10^9 and 5.2×10^9 parameters for budgets of 10^{19} to 10^{22} FLOPs, which is 0.45 decades of size per decade of compute, the slope a of panel h. Panel h also shows the cost of a fitting choice: the same runs give $a = 0.45$ under one fit and 0.51 under another, and at 10^{24} FLOPs those two lines are a factor of about two apart in N^*.
SRC: `experiments/041-scaling-laws/results/scaling_law_forms_summary.json, isoflop_optima_N, fits`

Panel i is the exercise that decides whether a curve is worth extrapolating. The pure power law fits the fifteen small runs well enough to pass by eye and gives $\alpha = 0.149$ with a 90% bootstrap interval of 0.134 to 0.164; it predicts losses of 1.60 and 1.27 at the two held-out sizes, where the observed means are 1.99 and 1.85, and its band at 10^{10} excludes every held-out point. The floored fit recovers $\alpha = 0.348$ against a generating value of 0.34, with an interval of 0.316 to 0.376, and $E = 1.71$ against 1.69, and predicts 1.96 and 1.86 at the held-out sizes. Nothing in the fitted range distinguishes the two models; the held-out runs do. That is the reason the largest runs are held out rather than fit.
SRC: `experiments/041-scaling-laws/results/scaling_law_forms_summary.json, synthetic_fits`

The animated versions. The Dendron note `notes/experiments.041-scaling-laws.scripts.scaling_law_forms.md` embeds nine GIFs the same script writes, one per panel and one sweep each, with the equation, the constants, the swept value and a one-sentence takeaway in every frame's title: α_N moving while the curve of panel a is pinned at the center of the fitted range so that only the forecast moves, α_D doing the same for panel b, E rising under a fixed $AN^{-\alpha}$ so the bend of panel c appears, one budget line sliding across the surface of panel d, D rising under the N-curve of panel e, N rising under the D-curve of panel f, C rising under the IsoFLOP curve of panel g with the minimum tracing the frontier, β moving so the allocation exponents of panel h move with it, and one bootstrap resample and refit per frame accumulating into the band of panel i.

3 Counting N, D and C when the examples are strains ✗

The forms were fit on language models, where the units are settled: a parameter is a weight, a datum is a token, and compute is $6ND$. For a model whose examples are genotype, environment and phenotype records, each unit has to be chosen, and a curve is only as defensible as the choice. The rule that resolves most of the cases is that D counts in the unit the loss averages over, and that the unit is held fixed across every run in the sweep.

D is instances consumed, not dimensions. D is the number of training examples the optimizer has processed when the loss is measured: the number of records in the training subset times the number of epochs, in the same unit the loss is averaged over. For a language model that unit is the token because the cross-entropy is per token; for a model trained on one loss per record it is the record. The dimension of an example, meaning the number of features in a record, the number of genes in the genome the model attends over, or the length of a sequence embedding, is not D. Dimension changes what a record is and how much compute each record costs, and it changes N through the width of the input layer, but a sweep over dimension is a sweep over architecture, not over data, and belongs on a different axis of the ablation. Two consequences follow.

- A record with several supervised labels, a genotype measured on several phenotypes, counts as one record under a summed loss and as one example per label under a per-label loss. Either is a valid unit; what breaks the curve is changing the unit between runs, or reporting a loss averaged over one unit against a D counted in another.
- A gene-token model that reads a genome as a set of tokens does not consume $D \times 6{,}000$ tokens in the Kaplan sense: the loss is still one number per record, and the tokens inside a record are the dimension, not the data. The per-record compute does scale with that token count, which is why C is measured rather than computed for such a model (below).

Repeated epochs count for less. Language models are trained for about one epoch, so D equals the corpus. A genotype-to-phenotype model trains for many epochs over a fixed corpus, and a repeated record is not worth a fresh one. Muennighoff et al. (arXiv:2305.16264)▲ [external: arXiv:2305.16264, not in the mirror] fit the Chinchilla form with an **effective data** term in which repeated tokens decay in value: up to about four epochs the loss is close to what the same number of fresh tokens would give, and beyond that the return on repetition falls off quickly. The usable statement for a data-limited project is that D in Equation 3 is the number of distinct records, and the epoch count is a training hyperparameter held fixed across the sweep, not a multiplier on D. A D-sweep is then a sweep over the size of the

distinct training set, which is what asks the question the project needs answered: whether a new experiment or a bigger model will pay more.

Subsample by the unit that leaks. A D-sweep draws nested subsets of the training set, 1/64, 1/16, 1/4 and the whole, against one fixed evaluation set. If test genotypes share genes or perturbations with training genotypes, adding data partly measures memorization of those shared parts, and the apparent exponent is inflated. The subsets are therefore drawn by the unit that leaks: by gene, or by the query pair a screen was built around, so that a smaller subset removes whole genes or whole screens rather than thinning records uniformly. The 025 additive-baselines document showed the size of this effect on the trigenic build, where the additive ridge falls from 0.400 test Pearson under a random-over-records split to 0.185 under a query-pair-disjoint one; a D-sweep drawn by record on that build would scale the first number, not the second.

N: count what the loss responds to. N is the number of trainable parameters that the loss responds to as capacity. Kaplan's exclusion of the embedding matrix has a direct analog: a gene-token model carries an embedding table with one row per gene, whose size is set by the genome rather than by the architecture, and that table should be reported separately from the rest, with the curve fit on the rest. Sizes are then produced by scaling width and depth together in a fixed ratio, five to seven of them spaced evenly in $\log N$ over at least one and a half to two decades.

C: measure it. The $6ND$ rule is transformer-specific and per token. For a model whose per-record cost depends on how many gene tokens a record carries, on graph message passing, or on a sequence encoder run once per gene, C is read from a profiler for one training step at each size and multiplied by the number of steps. `torch.utils.flop_counter.FlopCounterMode` does this for a PyTorch model without instrumentation. The IsoFLOP construction of panel g then uses the measured cost per record in place of $6N$.

E has a physical reading. For a language model E is the entropy of text and is not independently measurable. For a phenotype measured with replicates it is. Under a mean-squared loss the best possible predictor has expected error equal to the measurement-noise variance, so the replicate variance of the evaluation set is a lower bound on E in the loss's own units, and it is available before any model is trained. The released standard-deviation columns of the Costanzo and Kuzmin screens, whose replicate structure is recorded in `notes/torchcell.datasets.scerevisiae.costanzo2016.noise-computation.md`, give the per-record σ from which that bound is computed. A fitted E that falls below it says the fit is wrong, and one far above it says the model family is leaving something on the table that neither more data nor more parameters will recover. Which of those holds on the trigenic build has not been measured.

4 A minimal ablation, and what makes its curve defensible ✗

A Kaplan-style ablation on a modest allocation is three sweeps and four controls. The sweeps produce the points; the controls decide whether a line through them means anything.

The three sweeps.

1. **Parameters.** One architecture family, width and depth scaled in a fixed ratio, five to seven sizes spaced evenly in $\log N$ over at least one and a half to two decades, each trained on the full training set until the loss stops improving. This gives $L(N)$ with data not limiting, the row of panel a.
2. **Data.** One model large enough that data is the bottleneck, trained on nested subsets of the training set at 1/64, 1/16, 1/4 and 1, drawn by the unit that leaks (Section 3), against one fixed evaluation set. This gives $L(D)$, the row of panel b, and for a scientific dataset it is the sweep that matters most, since D cannot simply be increased.
3. **Compute.** Three or four budgets C. At each, several sizes N with D set by the budget, $D = C/c_N$ where c_N is the measured cost per record at that size. Each budget gives a U in N, its minimum is $N^*(C)$, and a line through the minima in log-log is the frontier of panel h. This is the Chinchilla IsoFLOP construction, and it is the only one of the three that answers how a fixed budget should be split.

The four controls.

- **A learning rate that suits each size.** Kaplan's runs used a cosine schedule whose length did not match each run's length, and the conclusion about N versus D changed when Hoffmann matched them. Either tune the learning rate per size, or use μP (Yang et al., arXiv:2203.03466)▲ [external: arXiv:2203.03466, not in the mirror], under which the optimum transfers across widths so one tuning at the smallest size serves the sweep.
- **Validation loss, not a downstream metric.** Loss is smooth in N and D; a correlation or an accuracy jumps and plateaus, and a power law fit to one is a fit to the plateau's edge.

- **Seeds at the small sizes.** Two or three seeds at each of the small sizes, where they are cheap, give the noise floor the bootstrap needs. Panel i uses three.
- **Compute counted properly.** $6ND$ for a transformer over tokens; a profiler for anything else.

Fitting. For a pure power law, a linear regression of $\log L$ on $\log N$ is the fit, and the slope is the exponent. With a floor the fit is nonlinear least squares in log space over $(\log A, \log E, \alpha)$, run from a grid of starting points and keeping the best, because the objective has several local minima; fitting the logarithms keeps A and E positive without constraints. The estimator in `experiments/041-scaling-laws/scripts/scaling_law_forms.py` (function `fit_floored`) does this with 27 starts and a squared residual. Hoffmann's appendix uses a Huber residual, which stops a few bad runs from steering the exponent; it is the right choice once real runs are being fit and is a one-line change there. The joint form adds $(\log B, \beta)$ to the parameter vector and is otherwise the same fit.

What makes the curve defensible.

- **Hold out the largest runs.** Fit on the smaller runs and check whether the fit predicts the largest. Panel i is the case where this is the only test that separates a right model from a wrong one.
- **Bootstrap over runs.** Resample runs with replacement, refit, and report an interval on every exponent. A point estimate of α is not a result, and panel h shows two published point estimates from one set of runs.
- **Look for curvature.** A bend at the large end means E is near, or that hyperparameters or data have become the bottleneck. Either way the straight line stops there.
- **Check E against the replicate floor.** When the loss is a squared error on a replicated measurement, the replicate variance is an estimate of E that owes nothing to the fit (Section 3).

5 What an ablation looks like on the trigenic build ✗

Measured. No number here is a torchcell result. The constants are the published ones and panel i is synthetic.
Not measured. Every number a scaling section of the manuscript or thesis would state: the exponents, the floor, and whether a curve fit on small runs predicts a large one. What follows is the design, with the choices each sweep forces written down so a run can be launched against it.

The axes available. The trigenic build of experiment 025 holds 376,732 records with bit-identical labels to the 010 build, organized as 420 recurring query pairs plus one array gene each, and it already carries the two splits an ablation needs: the published random-over-records split and a query-pair-disjoint split under which no query double is shared between parts. The equivariant cell graph transformer trained on it is one architecture family whose width and depth can be scaled together. The axes are therefore available now; what is missing is the sweep.

The data sweep first. Data is the binding axis here, so the D-sweep is the one to run before any other. Nested subsets of the disjoint-split training set at 1/64, 1/16, 1/4 and 1, drawn by query pair so that a smaller subset removes whole screens, against the one fixed disjoint test set; a model at the current configuration's size; the loss recorded per epoch and the validation-selected checkpoint scored on test. Four points give a slope and, with the held-out largest subset, the first extrapolation test. The reading the sweep is for: whether the loss is still falling with D at the full set, which says a new screen is worth more than a larger model, or has flattened, which says the opposite.

The parameter sweep. Five sizes of the transformer, width and depth in the fixed ratio of the current configuration, spaced evenly in $\log N$ over about two decades, with the gene embedding table counted separately and the curve fit on the rest; three seeds at the two smallest sizes; learning rate tuned per size, or transferred under μP once the parametrization is checked against the model code. Every run on the full disjoint training set, so that this sweep reads $L(N)$ with data as fixed as it can be made.

The floor from the replicates. Before either sweep, the replicate standard deviations of the test records give the noise-variance bound on E in the loss's units. Whether the trigenic interaction score's released uncertainty is a sample standard deviation over colonies or a bootstrap standard error, and over how many colonies, is already recorded for the Kuzmin and Costanzo screens in the noise-computation note, and the bound follows from it. That number is the first thing to set beside a fitted E.

Compute. The cost per record of the transformer depends on how many gene tokens a record carries and on the graph penalty, so C is read from a profiler at each size rather than from $6ND$. The IsoFLOP sweep is third in priority: it answers how to split a budget, which matters once the first two sweeps have shown that the budget buys anything.

What the section in the manuscript would then say. A scaling section is one figure, three panels, L against D, against N and against C, each with its fitted exponent and interval, the held-out largest run marked, and E drawn against the replicate floor. Until the sweeps have run, that figure is a plan, and the sweeps above are its specification.

6 What the expression and proteome rounds already pin down ✗

Measured. The replicate floor of every 019 panel in Pearson units; a budget curve over eight arms from 350 to 9,900 epochs; the disagreement between the validation loss and the metric in 27 long runs; a joint-training null on eleven partitions; the sequence-baseline bar and the best arm of the twelve-seed round.
Not measured. A parameter sweep and a data sweep on the expression store. Every 019 round varied one architectural choice at a fixed size on a fixed store, and the sizes that differ between rounds differ with the store and the input stack, so no exponent can be read from the record.

Experiment 019 trained the equivariant cell graph transformer on the single-deletion transcriptome (Kemmeren 2014), the single-deletion proteome (Messner 2023) and morphology (Ohya 2005), alone, jointly and conditioned on one another, across about twenty rounds. Read against Sections 3 and 4, that record settles the floor, exhibits the compute axis on the wrong statistic, and leaves both scaling axes unmeasured. Each piece is listed with what it is good for.

The floor, per panel. Table 3 is the measurement Section 3 asks for before any sweep, done for four panels. Kemmeren's two independent cultures per deletion give a test-retest ceiling of a mean Pearson per feature of 0.775 across strains; Caudal's 29 re-cultured isolates 0.604; Messner's duplicated-origin strains 0.405 against 0.696 from its wild-type replicates, two estimators that disagree because 146 pairs determine neither well; morphology 0.611 from 122 wild-type replicates. These are in Pearson, the campaign's metric. The bound on E in loss units needs the per-gene replicate variance, which Kemmeren's store carries as a standard error and a replicate count per value and which has not been converted; for the other panels only the pooled reliability is on file.

Table 3: The replicate ceiling of each 019 panel, the bound on E in Pearson units. Reliability is the per-feature fraction of across-strain variance that is signal (test-retest: the correlation of two independent measurements across strains; decomposition: one minus replicate variance over total variance), averaged over features; the ceiling is the mean of its square root, the best across-strain Pearson a perfect predictor of the signal can reach against a single noisy measurement. Two estimators are given where both exist because they disagree where few pairs determine either.

panel	estimator	reliability	ceiling, mean Pearson per feature
Kemmeren 2014, expression	test-retest, 82 paired deletions	0.611	0.775
	decomposition	–	0.862
Caudal 2024, expression	test-retest, 29 re-cultured isolates	0.405	0.604
	decomposition	0.235	0.459
Messner 2023, proteome	test-retest, 146 duplicated-origin ORFs	0.200	0.405
	decomposition, 388 wild-type replicates	0.501	0.696
Ohya 2005, morphology	decomposition, 122 wild-type replicates	0.444	0.611

The compute axis, in Pearson. Table 4 is a budget sweep of the kind panel c of the figure draws, read off the training curves of eight arms rather than run as one: the same runs rescored on the prefix of their own history at ten budgets. The mean rises from 0.120 at 350 epochs to 0.197 at 9,900, and flattens: the tenfold budget past 1,000 epochs buys 0.036. That is the shape a power law with a floor has, and it cannot be fit as one, because the statistic is a correlation and not a loss. The document's second control (Section 4) exists for exactly this case.

Why the loss cannot be the objective as it stands. In the 27 long runs of the same round whose loss is not the metric, the validation loss bottoms out at a median epoch 481 (the latest at 3,097) and the Pearson peaks at a median epoch 2,433, with the loss already 6.8 % above its minimum and +0.057 of Pearson gained between the two; the final loss sits 9.0 % above its minimum. The shrinkage probe on saved validation predictions says why: the expression predictions are 1.4 to 2.2 times too spread for their correlation on 3 checkpoints (the proteome's run the other way, 0.5 to 0.7 on 3), and one scalar per run brings the squared error below the per-gene mean. The loss is a scale problem riding on a correlation that keeps improving. A scaling fit on the raw loss would therefore read a calibration artifact as the curve. Two repairs are available and neither has been run as a sweep: fit on a calibrated loss (the squared error after the one-scalar shrinkage, which is the probe's statistic), or fit on the per-gene Pearson with the understanding that it is bounded and will bend near the floor for a reason that is not E.

The bar a curve has to start above. On the store the twelve-seed round trains on, the nearest-neighbor average over ProtT5 embeddings reaches 0.110 ± 0.017 validation Pearson per feature over 12 split seeds. The best arm of

Table 4: The compute axis as the 019 campaign measured it: the same eight long-budget arms of the v9 round rescored on the prefix of their own curves at each budget (the maximum of a centered five-epoch rolling mean of the validation Pearson per feature over epochs up to the budget). Mean and sd over the eight arms; the spread is an arm spread plus nondeterminism, so it bounds the replicate spread from above. The statistic is Pearson, not the loss, for the reason given in the text.

epochs	arms	mean Pearson per feature	sd	min	max
350	8	0.1198	0.0096	0.1029	0.1317
500	8	0.1223	0.0058	0.1130	0.1317
700	8	0.1411	0.0166	0.1207	0.1606
1,000	8	0.1609	0.0099	0.1485	0.1780
1,500	8	0.1670	0.0117	0.1536	0.1839
2,000	8	0.1745	0.0151	0.1547	0.1950
2,800	8	0.1821	0.0175	0.1629	0.2085
4,000	8	0.1883	0.0171	0.1663	0.2145
6,000	8	0.1917	0.0177	0.1663	0.2222
9,900	8	0.1965	0.0222	0.1663	0.2382

that round, `S_mask`, reaches 0.094 ± 0.028 over twelve seeds, and the reference 0.063 ± 0.037. Morphology realizes 13 % of its ceiling. A curve fit to a model that sits at or below a trivial predictor measures the predictor, so on this store the first point of either sweep has to clear the neighbor average before the second point means anything. On the larger expression store the v13 reference sits within 0.01 of the same baseline (Figure 3 gate document, branch `multimodal-phenotype-retrospective`).

The multimodal data axis. Adding the proteome's strains and labels to the trunk is a move along D on a second modality, and the joint round measured it on eleven partitions: the joint arm against the expression-only arm -0.0012 (5 of 11 partitions above zero, reference 0.094), and against the proteome-only arm -0.0172 (1 of 11 above zero, against a non-inferiority margin of 0.015). Pooling a second modality did not move the curve. Conditioning did, by +0.03 to +0.05 over the unconditioned arm (Figure 3 gate document), and stayed below a ridge from the measured modality, which is the linear baseline for that axis. This is the one data-axis measurement the record holds, and it says the second modality is worth more as an input than as more training signal for a shared trunk.

What the record says about N, and why it is not a curve. The expression strand's own summary reads 145 runs from 0.12M to 4.7M parameters with a correlation of -0.129 between $\log_{10} N$ and score (`notes-tex/019-simb-multimodal-expression/sections/0-strand.tex`), and the Figure 3 gate found that no committed script reproduces that subset while all 1,182 scored runs give -0.001. Either way the runs differ in budget, input stack, head and objective at once, so the reading is a confounded null, not $L(N)$. The one deliberate width increase on the fast code, four layers at width 180 against six at width 90, collapsed to the per-gene-mean predictor on 2 of 2 seeds, which is the first control of Section 4 failing before the sweep could start: the learning rate that suits width 90 does not suit width 180.

What the record does not say about D. The best per-gene scores of the campaign, the v13 reference at a window mean of 0.20 and 0.17 on split seeds 0 and 1 (`v13\_split\_readout.json`), were trained on about the same number of strains as the twelve-seed round (the fig3_core expression rows against the 1,103 both-label rows), with the full four-embedding input stack and a 6,000-epoch budget. The gap to the small trunk's 0.06 to 0.10 is therefore budget and input, not D, and the record holds no two points that differ in D alone.

The two sweeps, launched on cabbi 2026-10-09. Both run on the fast recipe with the mask schedule on (the one twelve-seed arm with zero collapses) at batch 128 and the unscaled rate, one run per RTX6000 card, three split seeds, W&B project `torchcell\_019\_scaling`, stages `scale\_d` and `scale\_n` of `igb\_expr\_wave5.slurm` (IGB jobs 2427637 and 2427638). The statistic is still the per-gene Pearson at the registered window, with the loss curve logged for the calibrated fit later.

- **Data.** Nested prefixes of one split-seeded permutation of the 1,103 both-label training rows at 1, 1/2, 1/4, 1/8 and 1/16, validation and test untouched (`train\_cgt\_multitask.py`, `train_fraction`). The budget is 1,200 epochs for every fraction, so the small fractions take fewer optimizer steps (one per epoch at 1/16); the per-epoch validation curve is what the readout reads and the step mismatch is part of the result, not hidden by it.
- **Parameters.** Widths 45, 90, 180 and 360 at six layers (nine heads divide all four), the learning rate scaled as 1/width from 3×10^{-4} at 90. Hypothesis (untested): that rule keeps the wider trunks out of the collapse the width-180 arm fell into at the width-90 rate; it is not μP, and a width whose rate is wrong will show as a collapse rather than a point on the curve. Width 360 is last in the array so an out-of-memory there costs the sweep nothing else.

7 Sources ✗

The first three works below are in the mirror as of 2026-10-11, under the citation keys given, so their numbers carry no external flag. The last two are not, and are flagged where their claims are used. The document has no Zotero collection yet, so every work is named in prose; once a collection exists, the mirrored keys become `\cite` keys.

- Kaplan et al., 2020, *Scaling Laws for Neural Language Models*, arXiv:2001.08361.
  Mirror key `kaplanScalingLawsNeural2020`.
  The single-axis forms of Equation 1, the non-embedding parameter count, and the $6ND$ rule.
- Hoffmann et al., 2022, *Training Compute-Optimal Large Language Models*, arXiv:2203.15556.
  Mirror key `hoffmannTrainingComputeOptimalLarge2022`.
  The joint form with a floor, the three fitting approaches (its appendix covers all three, including the Huber residual), the IsoFLOP construction, and the Approach 3 constants.
- Besiroglu et al., 2024, *Chinchilla Scaling: A Replication Attempt*, arXiv:2404.10102.
  Mirror key `besirogluChinchillaScalingReplication2024`.
  The refit of the Approach 3 data and the second set of constants in Table 1.
- Yang et al., 2022, *Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer*, arXiv:2203.03466. μP, the parametrization under which the learning-rate optimum transfers across widths.
- Muennighoff et al., 2023, *Scaling Data-Constrained Language Models*, arXiv:2305.16264. The effective-data form for repeated epochs.

Within the repository: the 025 additive-baselines document (`notes-tex/multimodal/025-additive-baselines/`) for the record count, the splits and the leakage measurement quoted in Section 3, and `notes/torchcell.datasets.scerevisiae.costanzo2016.noise-computation.md` for the replicate structure behind the floor estimate.